

Volatility: The Open Source Framework for Memory Forensics

Computer attacks are a constant concern for admins and users of computers. These are attacks that are stealthy enough not to leave any traces on the hard disk of the computer. To detect such attacks, we need to make a forensic analysis of the memory dump of the computer. This analysis is termed memory forensics. Volatility is the open source framework that could help us with memory forensics.



According to Wikipedia, “Memory analysis is the science of using a memory image to get information about running programs, the operating system, and the overall state of a computer.” Volatile memory contains valuable information about the runtime state of the system (the network, file system and registry). In this article, we are going to investigate the digital artifacts of volatile memory using Volatility.

Introducing Volatility

Volatility is an open source framework used for memory forensics and digital investigations. The framework inspects and extracts the memory artifacts of both 32-bit and 64-bit systems. The framework has support for all flavours of Linux, Windows, MacOS and Android. It can analyse raw memory dumps, crash dumps, virtual machine snapshots, VMware dumps (*.vmem*), Microsoft crash dumps, hibernation files, virtual box dumps, and many others. The framework is intended to investigate the system's state independently and consists of over 35 plugins for analysing. Volatility is available as pre-installed binaries in several Linux flavours such as REMnux, Kali

Linux, etc. It is freely available in Git (<https://github.com/volatilityfoundation/volatility>).

Memory format support

Volatility supports a variety of sample file formats and has the ability to convert between these formats. These are:

- Raw/padded physical memory
- Firewire (IEEE 1394)
- Expert Witness (EWF)
- 32-bit and 64-bit Windows Crash Dump
- 32-bit and 64-bit Windows Hibernation File
- 32-bit and 64-bit Mach-O files
- Virtualbox ELF64 Core Dumps
- VMware Saved State (*.vmsx*) and Snapshot (*.vmsn*)
- HPAK format (FastDump/FDpro)
- QEMU memory dumps
- LiME (Linux Memory Extractor) format
- Mac-O file format

Operating system support

- Support for all 32-bit and 64-bit Windows systems
- Support for 32-bit and 64-bit Linux kernels ≤ 4.2